OPTICROP: SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE

INTERNSHIP PROJECT

PROJECT USER & SYSTEM MANUAL

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: https://github.com/sribalakatta/Project-OptiCrop-Smart-Agricultural-Production-Optimization-Engine-

1. Source Repository & Cloning

The official source repository is hosted on GitHub. Use the link below to view or clone the code assets:

- GitHub Link: https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine
- Clone command: git clone https://github.com/sribalakatta/Project-OptiCrop-Smart-Agricultural-Production-Optimization-Engine-

2. Setup Instructions

Follow these steps to run the OptiCrop engine on a local machine:

- Prerequisites: Install Python 3.10+ and git version control.
- Setup Environment: Run 'python -m venv .venv' and activate it.
- Install Packages: Run 'pip install -r requirements.txt'.
- Data & Models: Run 'python generate_datasets.py' and 'python train_models.py' to generate model files.
- Run Server: Run 'python app.py' and open 'http://localhost:5000' in a web browser.

3. Operating Instructions

Enter NPK and environmental values in the form, then click 'Execute Production Optimization'. Review the suggested crop and fertilizer recommendations.